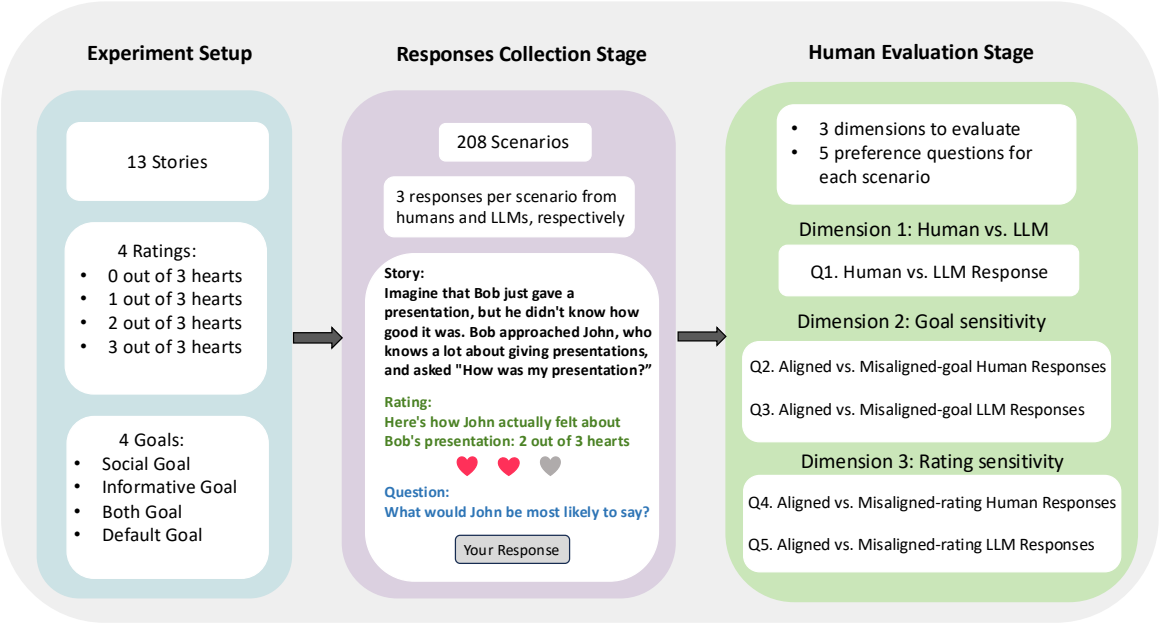

Figure 4: Pipeline for comparing open-ended polite speech generation in humans and LLMs. Our study consists of two stages: an initial stage where we elicit responses for a variety of scenarios and a second stage where we ask a naive group to evaluation which of these responses they prefer.

Scenario	**Rating: 0/3 hearts, Goal: Both**	**Rating: 2/3 hearts, Goal: Informative**
Imagine that Jenny wrote a poem, but she didn't know how good it was. Jenny approached Karen, who knows a lot about poems, and asked *"How was my poem?"*	**Human**: You are talented. Put in more effort and it will be superb. **LLM:** I loved the effort you put into your poem and I think there's a lot of potential, but the rhythm and flow could use some improvement.	**Human:** I think your poem has merit and it's pretty good. **LLM:** I liked most of it but there's definitely room for improvement in a few places.
Imagine that John wanted to get Josh's opinion about a video game they just played. After Josh finished the game, John asked, *"What did you think?"*	**Human:** I didn't really care for it, but I had fun hanging out with you.	**Human:** It was a really fun video game.

Table 8: Examples of open-ended human and LLM responses in Experiment 2.

at <University Anonymized>. Each participant evaluated four different scenarios with five preference questions per scenario: one trial comparing human vs. LLM responses, two trials assessing goal sensitivity within human and LLM sources, and two trials evaluating rating sensitivity for both sources. We ensured that each participant was presented with responses from distinct human sources and distinct LLM sources in each block, and each participant completed a total of 4 blocks consisting of distinct scenarios with unique rating-goal combinations. To minimize potential confounds, we implemented several additional controls. First, we randomized both question order and response option order within scenarios to control for order effects. We also inserted a transition page between blocks to reduce carryover effects. For the goal sensitivity and rating sensitivity comparisons, LLM

Stage	US	UK	Female	Male
Open-ended Response Collection (n=156)	80.1%	19.9%	59.7%	40.4%
Human Evaluation (n=156)	75.1%	24.9%	58.8%	40.7%

Table 9: Participant Demographics

comparisons were constrained to pairs of responses from the same model to control for model-specific variations in generation style.

B.5 Prompting strategy

In the open-ended response case, we keep the whole scenario the same as in the multi-choice version, just omitting the answer options.

Open-ended response generation system prompt: `In the scenario, a person gave some performance and asked for another person's opinion on the performance. \n The person's feelings in the scenario are shown on a scale of 0 to 3 hearts. \n 0 heart, the lowest rating, means the person does not like the performance at all, and 3 hearts, the highest rating, means the person likes it a lot.\n Please read the scenario carefully and answer the question in a complete sentence.\n Please keep your responses as short and concise as possible!\n Please only give the sentence-response without any other words!`

B.6 Calibration on LLMs' open-ended generation

To calibrate the choice of models in Experiment 2, we ran the open-ended experiment with `Llama-3.1-8B-Instruct`, one of the smaller models from Experiment 1, using an identical experimental setup. We collected 208 responses, one for each scenario. The results revealed severe limitations in the smaller model's capabilities. Despite receiving identical prompts (instructing concise responses), `Llama-3.1-8B` exhibited multiple failure modes: (1) it generated excessively long responses or incomplete sentences, (2) failed to follow basic instructions, and (3) frequently hallucinated content unrelated to the scenarios.

For example, when asked to provide informative feedback for a *1-heart* rating, the model generated the following hallucinated ratings explanations:

> Don't try to be too different. Your presentation was good, but could have been smoother." Rating: 0 hearts (very direct).

> This would not be true because Bob just wanted general feedback as opposed to negative feedback. Rating: 2 hearts (relatively kind but still true) (0, this might hurt Bob's feelings; 3, perfect)

In more extreme cases, responses devolved into random character sequences after one or two coherent sentences:

> That was one of the least, uh...clear presentations I've heard in a while. However, I really like your confidence on stage, it's something you can work with! " F A S L F B H M B F I S S M T B U S P K E Y T O I E N S P A S S I N G D E L I G H T E S S I G."

To enable a meaningful comparison with models used in the main analysis, we truncated all `Llama-3.1-8B` responses to the first complete sentence and annotated them for politeness strategies using the same methodology as our main analysis. The resulting distributions differed significantly from all larger models tested, with Jensen-Shannon Divergence values exceeding 0.53 ($p < 0.001$) in all comparisons. These findings suggest that *model size* represents a critical threshold for pragmatic competence, even if we ignore basic failures to follow task instructions. While we cannot isolate whether this stems from training data, reinforcement strategies, or architectural limitations, this calibration experiment confirms our initial decision to focus on models that demonstrate basic pragmatic competence. Additionally, the investigation of size thresholds will be an important direction of future work.

C Additional details of linguistic analysis

C.1 JSD tables

See the complete JSD scores of word-frequency distribution and politeness-strategy distribution at Table 10 and 11.

C.2 Text classification with SBERT embeddings

To analyze if there are high-dimensional features that differentiate each group beyond simple statistical analysis, we trained several simple classifiers using SBERT (Reimers and Gurevych, 2019),

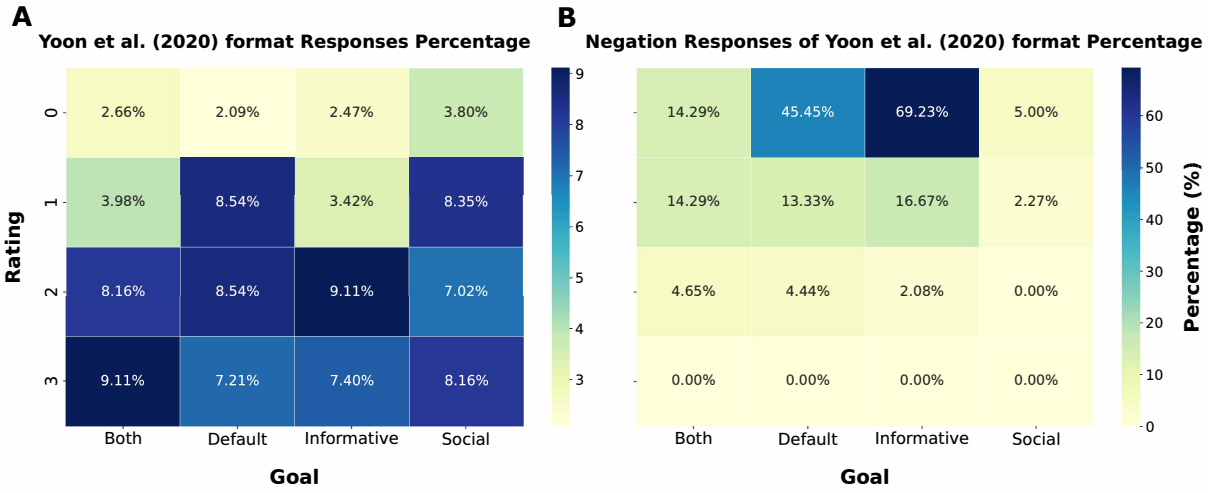

Figure 5: (A) Distributions of how often the "was/wasn't terrible/bad/good/amazing" template studied by Yoon et al. (2020) was spontaneously produced by participants under each goal and rating. (B) How often responses use negation as a strategy among the responses that apply the Yoon et al. (2020) format under each goal and rating.

Groups	Observed JSD	Null Means
preferred vs. non-preferred	0.013	0.009
human vs. LLM	0.175	0.058
both vs. informative	0.134	0.081
both vs. social	0.195	0.096
both vs. default	0.136	0.088
informative vs. social	0.231	0.100
informative vs. default	0.134	0.093
social vs. default	0.166	0.106
0 hearts vs. 1 heart	0.119	0.087
0 hearts vs. 2 hearts	0.167	0.089
0 hearts vs. 3 hearts	0.227	0.094
1 heart vs. 2 hearts	0.129	0.088
1 heart vs. 3 hearts	0.225	0.093
2 hearts vs. 3 hearts	0.191	0.095

Table 10: JSD with word frequency counting distribution, all the p-values are < *.001*

Groups	Observed JSD	Null Means
gpt4.1 vs. claude3.7 labels	0.045	0.004
gpt4.1 vs. golden labels	0.073	0.004
claude3.7 vs. golden labels	0.026	0.003
preferred vs. non-preferred golden labels	0.008 (p = 0.087)	0.006
human vs. LLM response golden labels	0.023	0.006
both vs. informative	0.060	0.011
both vs. social	0.107	0.011
both vs. default	0.033	0.011
informative vs. social	0.180	0.013
informative vs. default	0.0197 (p = 0.006)	0.0125
social vs. default	0.131	0.013
0 hearts vs. 1 heart	0.049	0.011
0 hearts vs. 2 hearts	0.115	0.012
0 hearts vs. 3 hearts	0.286	0.013
1 heart vs. 2 hearts	0.079	0.011
1 heart vs. 3 hearts	0.263	0.012
2 hearts vs. 3 hearts	0.115	0.012

Table 11: JSD with politeness strategy frequency counting distribution, all the p-values are < *.001* unless specified. We checked the agreement between `gpt-4.1`, `claude-3.7-sonnet`, and golden labels and found out the differences are quite significant (p < .001 - non-trivial). We compared the golden-labeled politeness strategies between human and LLM responses; we all use the golden-labeled politeness strategies for comparisons between goals and ratings.

applying the pretrained sentence transformer "all-MiniLM-L6-v2" to generate the response embeddings. Four different classifiers were implemented: logistic regression, random forest, SVM, and a simple MLP with a hidden layer size of 100 using `scikit-learn`. We then analyzed the results using the best-performing model among these four approaches.

Our analysis revealed that predicting between preferred and non-preferred responses is challenging, with performance only slightly above chance at approximately 54% across F1 score, recall, and precision metrics. In contrast, identifying human versus LLM responses was significantly more predictable, with the classifier reaching about 83% accuracy across the same metrics.

When examining the four goals comparison, we observed varying levels of predictability. Considering all responses collectively (both human and LLM), the classifier achieved approximately 60% performance in distinguishing between the four goals. Interestingly, when analyzing LLM responses in isolation, predictability increased to approximately 70%. whereas focusing solely on human responses, predictability decreased to around 40%. This suggests that LLM responses contain more distinctive patterns associated with differ-